Response to the Editor

Manuscript PONE-D-26-28965R1

Zero-shot generalization of EEG attention-state decoding across increasingly stringent validation strategies on open datasets

Dear Dr. Fathi Jouzdani,

[TODO — opening paragraph and summary of the revision.]

Your comments are reproduced below and followed by our point-by-point responses, each indicating where in the revised manuscript the issue has been addressed.

Yours sincerely,

Sanzhar Sovet, Beibit Abdikenov, Manzura Zholdasova, Altyngul Kamzanova, Medet Mukushev and Almira Kustubayeva

Additional Editor Comments

Comment 1

Preprocessing and data leakage: Please specify exactly which data are used to fit ICA and all other data-dependent preprocessing steps in each validation protocol. Clarify whether any information from held-out participants or target datasets contributes to preprocessing, feature transformation, model selection, or other fitted operations. Explain how the protocol supports the claim of zero-shot generalization without target-specific adaptation.

Response:

[TODO — response.]

Comment 2

Uncertainty and sample size: Please discuss the limitations imposed by the five participants in Dataset A. Clarify the unit of resampling used for confidence intervals and ensure that the uncertainty estimates reflect the independent sampling units. Avoid interpreting small differences between baseline and cross-subject estimates as evidence of equivalence.

Response:

[TODO — response.]

Comment 3

Comparability of validation protocols: Please explain the extent to which the validation comparisons isolate changes in participant identity, task composition, and dataset. Ensure that transfer results are compared with appropriate target-specific baselines and that the conclusions do not attribute observed differences to a single factor when several factors change simultaneously.

Response:

[TODO — response.]

Comment 4

Interpretation of attention labels: Please maintain a clear distinction between classifying the protocol-defined conditions and decoding attention as a general psychological construct. Discuss the potential influence of task demands and other differences between resting and active cognitive conditions.

Response:

[TODO — response.]

Comment 5

Ethics and reporting: Please clarify the ethical approval, consent, and exemption status applicable to the original datasets and the present secondary analysis, as appropriate. Ensure that the ethics statement is consistent with the submission metadata.

Response:

[TODO — response.]

Comment 6

Final consistency check: Please verify that the final clean manuscript, figures, supplementary tables, and response letter contain consistent terminology, numerical results, and methodological descriptions.

Response:

[TODO — response.]